

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

**End-to-End Simulation of a Four-Element UHF Radio Interferometer for
Solar Burst Observation at 610 MHz**

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Electronics and Telecommunications Engineering (Bachelor of Science)

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neering

HANOI, 06/2026

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG ĐIỆN – ĐIỆN TỬ
ĐÁNH GIÁ ĐỒ ÁN TỐT NGHIỆP
(DÀNH CHO CÁN BỘ HƯỚNG DẪN)

Tên đề tài: End-to-End Simulation of a Four-Element UHF Radio Interferometer for Solar Burst Observation at 610 MHz

Họ tên SV: Nguyễn Bá Quân

MSSV: 20224350

Cán bộ hướng dẫn 1:

Cán bộ hướng dẫn 2:

STT	Tiêu chí (Điểm tối đa)	Hướng dẫn đánh giá tiêu chí	Điểm tiêu chí
1	Thái độ làm việc (2,5 điểm)	Nghiêm túc, tích cực và chủ động trong quá trình làm ĐATN Hoàn thành đầy đủ và đúng tiến độ các nội dung được GVHD giao	
2	Kỹ năng viết quyền ĐATN (2 điểm)	Trình bày đúng mẫu quy định, bố cục các chương logic và hợp lý: bảng biểu, hình ảnh rõ ràng, có tiêu đề, được đánh số thứ tự và được giải thích hay đề cập đến trong đồ án, có căn lề, dấu cách sau dấu chấm, dấu phẩy, có mở đầu chương và kết luận chương, có liệt kê tài liệu tham khảo và có trích dẫn, v.v. Kỹ năng diễn đạt, phân tích, giải thích, lập luận: cấu trúc câu rõ ràng, văn phong khoa học, lập luận logic và có cơ sở, thuật ngữ chuyên ngành phù hợp, v.v.	
3	Nội dung và kết quả đạt được (5 điểm)	Nêu rõ tính cấp thiết, ý nghĩa khoa học và thực tiễn của đề tài, các vấn đề và các giả thuyết, phạm vi ứng dụng của đề tài. Thực hiện đầy đủ quy trình nghiên cứu: đặt vấn đề, mục tiêu đề ra, phương pháp nghiên cứu/giải quyết vấn đề, kết quả đạt được, đánh giá và kết luận. Nội dung và kết quả được trình bày một cách logic và hợp lý, được phân tích và đánh giá thỏa đáng. Biện luận phân tích kết quả mô phỏng/phần mềm/thực nghiệm, so sánh kết quả đạt được với kết quả trước đó có liên quan. Chỉ rõ sự phù hợp giữa kết quả đạt được và mục tiêu ban đầu đề ra, đồng thời cung cấp lập luận để đề xuất hướng giải quyết có thể thực hiện trong tương lai. Hàm lượng khoa học/độ phức tạp cao, có tính mới/tính sáng tạo trong nội dung và kết quả đồ án.	
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Điểm tổng các tiêu chí:			
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Điểm từng tiêu chí cho lẻ đến 0,5. Nếu Điểm tổng các tiêu chí > 10 thì Điểm hướng dẫn làm tròn thành 10.

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(DÀNH CHO CÁN BỘ THÀNH VIÊN HỘI ĐỒNG)

Hội đồng số:

Họ tên SV: Nguyễn Bá Quân

MSSV: 20224350

Cán bộ thành viên HĐ:

STT	Tiêu chí (Điểm tối đa)	Hướng dẫn đánh giá tiêu chí	Điểm tiêu chí
1	Chất lượng slides/Bản vẽ kỹ thuật (1,5 điểm)	Sử dụng các minh họa hỗ trợ: hình ảnh, biểu đồ rõ nét và phù hợp, dễ hiểu	
		Không quá nhiều từ, biết sử dụng từ khoá; bố cục logic, có đánh số trang	
2	Kỹ năng thuyết trình (1,5 điểm)	Tự tin, làm chủ nội dung trình bày, đúng thời gian quy định	
		Dễ hiểu, dễ theo dõi, lô-gic, lôi cuốn.	
3	Nội dung và kết quả đạt được (5 điểm)	Nêu rõ tính cấp thiết, ý nghĩa khoa học và thực tiễn của đề tài, các vấn đề và các giả thuyết, phạm vi ứng dụng của đề tài. Thực hiện đầy đủ quy trình nghiên cứu: đặt vấn đề, mục tiêu đề ra, phương pháp nghiên cứu/giải quyết vấn đề, kết quả đạt được, đánh giá và kết luận.	
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		Chỉ rõ sự phù hợp giữa kết quả đạt được và mục tiêu ban đầu đề ra, đồng thời cung cấp lập luận để đề xuất hướng giải quyết có thể thực hiện trong tương lai. Hàm lượng khoa học/độ phức tạp cao, có tính mới/tính sáng tạo trong nội dung và kết quả đề án.	
4	Trả lời câu hỏi (2,5 điểm)	Trả lời ngắn gọn, chính xác, đi thẳng vào vấn đề của câu hỏi.	
		Nắm vững kiến thức cơ bản liên quan đến lĩnh vực nghiên cứu/công việc của đề án.	
5	Điểm thành tích (1 điểm)	Có bài báo KH được đăng hoặc chấp nhận đăng / đạt giải SV NCKH giải 3 cấp Trường trở lên / các giải thưởng khoa học trong nước, quốc tế từ giải 3 trở lên / có đăng ký bằng phát minh sáng chế. (1 điểm)	
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Điểm tổng các tiêu chí:			
Điểm đánh giá:			

Cán bộ thành viên hội đồng
(Ký và ghi rõ họ tên)

Điểm từng tiêu chí cho lẻ đến 0,5. Nếu Điểm tổng các tiêu chí > 10 thì Điểm đánh giá làm tròn thành 10.

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		Kỹ năng diễn đạt, phân tích, giải thích, lập luận: cấu trúc câu rõ ràng, văn phong khoa học, lập luận logic và có cơ sở, thuật ngữ chuyên ngành phù hợp, v.v.	
2	Nội dung và kết quả đạt được (5,5 điểm)	Nêu rõ tính cấp thiết, ý nghĩa khoa học và thực tiễn của đề tài, các vấn đề và các giả thuyết, phạm vi ứng dụng của đề tài. Thực hiện đầy đủ quy trình nghiên cứu: đặt vấn đề, mục tiêu đề ra, phương pháp nghiên cứu/giải quyết vấn đề, kết quả đạt được, đánh giá và kết luận.	
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Điểm tổng các tiêu chí:			
Điểm phản biện:			

Cán bộ phản biện
(Ký và ghi rõ họ tên)

Điểm từng tiêu chí cho lẻ đến 0,5. Nếu Điểm tổng các tiêu chí > 10 thì Điểm phản biện làm tròn thành 10.

ACKNOWLEDGMENT

Completing this graduation thesis marks the end of four years at Hanoi University of Science and Technology. For me, it is a wonderful experience of learning and researching in a creative and competitive environment.

But to have me today, it is the sacrifice and companionship of my parents, who have always cheered and encouraged me and been by my side, and I am always proud of them. I am grateful and thankful to my parents; no words can express this deep gratitude.

Moreover, I am profoundly grateful to all the teachers who imparted their knowledge and wisdom. I especially want to thank Mr. Han Huy Dung and Mr. Ha Duyen Trung for their patient guidance and the many hours of technical discussions in the SPARC Lab. Mr. Han Huy Dung helped me so much in stepping into the world of radio astronomy, along with Professor Vo Bich Hien, one of his colleagues in this field. Bringing me into the lab gave me the opportunity to work with the members of the SPARC Laboratory, set up real-time solar radio burst detection antennas, and access the measurement hardware that motivated this simulation work. I consider myself incredibly fortunate to have had their mentorship, guiding me through the challenges and triumphs of my studies.

Finally, I want to express my commitment to myself: to always stay focused and never give up, even in the most difficult areas of the telecommunication engineering major. Hanoi University of Science and Technology will forever be my second home—the place that helped me grow into a real telecommunication engineer, especially in the field of radio astronomy.

Student

(Signature and full name)

Nguyen Ba Quan

ABSTRACT

Currently, deploying real-time radio astronomy systems for educational and amateur purposes presents significant practical challenges. The construction of physical interferometry arrays demands high financial investment, high-performance computing hardware to process massive continuous data streams, and complex signal processing pipelines. Amateur radio astronomy engineers frequently suffer from hardware setup errors, synchronization issues, and software integration bottlenecks, making them highly inaccessible for small teams or educational institutions. Consequently, I design a software simulation—not only to verify and troubleshoot the design of physical systems prior to costly deployments but also to serve as an accessible educational platform.

This thesis presents the design and a complete, physically grounded software simulation of a four-element ultra-high-frequency (UHF) radio interferometer for observing the Sun at 610 MHz. The motivation is to validate, in a controlled and reproducible numerical environment, a small east–west baseline array of Yagi–Uda antennas of the kind that can be built at low cost for educational solar radio astronomy, and to demonstrate that such an array reproduces the classical signatures of aperture-synthesis imaging.

The work proceeds in three coupled stages. First, the receiving element—a commercial UHF Yagi–Uda antenna with a folded-dipole driven element on a four-bar U-frame reflector—is modelled in 4NEC2 and cross-validated against direct measurement and an independent finite-element solution. The simulated element exhibits an input impedance of $Z_{\text{in}} \approx 108 - j22 \, \Omega$, a forward gain of about 9.5 dBi, a half-power beamwidth near 52° , and a standing-wave ratio of 1.55 at 75 Ω ; mutual coupling between elements is found to be negligible across the operational baselines (falling below -78 dB beyond 5 m), confirming that the elements behave as independent sensors. Second, an end-to-end signal-processing pipeline is built in Python. Real archived solar flux from the SWS Learmonth observatory is converted to complex antenna voltages through the Friis reception relation with additive Johnson–Nyquist thermal noise; per-antenna geometric phases are applied using a standard local-to-equatorial $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation; the signals are cross-correlated to form visibilities; and the visibilities are inverted by a vectorised discrete Fourier transform to a dirty image. Third, the simulated instrument is exercised over the transit real time data at Learmonth station in Australia and over a sweep of array geometries.

The simulated interferometer reproduces the expected analytic behaviour: an ~ 8 -minute fringe period for the 15 m baseline, a jinc-shaped visibility decay as a function of baseline length consistent with a resolved solar disk, and a dirty image equal to the true sky convolved with the synthesised point-spread function. These results confirm that the proposed low-cost array is a sound instrument concept and that the simulation faithfully captures the physics from incident flux to final image, providing a validated design and analysis tool for solar radio interferometry.

Keywords: radio interferometry, aperture synthesis, solar radio bursts, Yagi–Uda antenna, 4NEC2, van Cittert–Zernike theorem, dirty image, point-spread function.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
ADC	Analog-to-Digital Converter
CST	Computer Simulation Technology (CST Studio Suite)
DFT	Discrete Fourier Transform
ENU	East–North–Up local coordinate frame
FFT	Fast Fourier Transform
FoV	Field of View
FWHM	Full Width at Half Maximum
HPBW	Half-Power Beam Width
IDG	Image-Domain Gridding
LNA	Low-Noise Amplifier
MoM	Method of Moments
NEC	Numerical Electromagnetics Code
PSF	Point-Spread Function (synthesised / dirty beam)
RF	Radio Frequency
RSTN	Radio Solar Telescope Network
SEFD	System-Equivalent Flux Density
sfu	Solar Flux Unit ($1 \times 10^{-22} \text{ W}/(\text{m}^2 \text{ Hz})$)
SNR	Signal-to-Noise Ratio
SWR	Standing-Wave Ratio
SWS	Space Weather Services (Australian Bureau of Meteorology)
TMS	Thompson, Moran & Swenson (standard interferometry reference)
UHF	Ultra-High Frequency (300 MHz–3 GHz)
UTC	Coordinated Universal Time
uv-plane	Spatial-frequency plane of the visibility function
VCZ	Van Cittert–Zernike (theorem)

CHAPTER 1. INTRODUCTION

1.1 Background and Motivation

The Sun is, after the Earth, the brightest and most variable radio source in the sky. Its radio output is generated by physical processes that span the full extent of the solar atmosphere, from thermal bremsstrahlung in the quiet corona to intense, short-lived bursts driven by particle acceleration during flares and coronal mass ejections. Monitoring this emission at metre and decimetre wavelengths is one of the oldest branches of radio astronomy and remains central to space-weather forecasting, because solar radio bursts are early, ground-observable signatures of the eruptive events that disturb the near-Earth environment.

A single dish, however, cannot simultaneously achieve the angular resolution and the collecting area required to localise a compact radio burst on the solar disk. The angular resolution of an aperture of diameter D at wavelength λ is of order λ/D ; at 610 MHz ($\lambda \approx 0.49$ m) a resolution comparable to the 32' solar diameter would demand a single antenna tens of metres across. *Radio interferometry* circumvents this limit by correlating the signals from several small, widely separated antennas: the achievable resolution is then set by the largest separation (the *baseline*) rather than by the size of any one element. This is the principle behind every modern synthesis array, from the historical Culgoora radioheliograph to today's LOFAR and the Murchison Widefield Array.

The motivation for this thesis is both pedagogical and practical. A small east–west interferometer built from inexpensive commercial UHF Yagi–Uda antennas is an attractive instrument for university-level solar radio astronomy: the hardware is affordable, the 610 MHz band sits in a relatively quiet portion of the UHF spectrum, and the geometry is simple enough to be understood analytically. Before committing to field measurements, however, it is essential to have a faithful, physically grounded *software model* of the complete instrument—from the incident solar flux, through the antennas and receiver chain, to the final synthesised image. Such a model makes it possible to predict the instrument's response, to verify the signal-processing chain against known analytic results, and to explore array geometries that would be costly to test in hardware. Building and validating that end-to-end simulation is the subject of this work.

1.2 Problem Statement

Constructing a working radio interferometer—even a modest four-element one—couples together several sub-problems that are usually treated in isolation in undergraduate courses:

- the **electromagnetic behaviour of the receiving element**: a real Yagi–Uda antenna has a frequency-dependent input impedance, a directional radiation pattern that acts as a *primary*

beam, and a mutual coupling to its neighbours that must be quantified before the elements can be treated as independent sensors;

- the **conversion of an astronomical flux density into a receiver signal**: the weak solar signal must be combined with the unavoidable thermal noise of the receiver in a statistically correct way;
- the **geometry of the array**: the time-varying delay between antennas as the Sun moves across the sky must be computed exactly, in three dimensions, for an arbitrary array layout and observing site;
- the **formation of the image**: the sparse set of correlated measurements (*visibilities*) must be transformed into a brightness (*dirty*) image, which is the true sky brightness convolved with the array’s point-spread function.

Each sub-problem is well understood individually, but assembling them into a single, self-consistent simulation—and demonstrating that the assembled instrument reproduces the analytic predictions of interferometry theory—is non-trivial. The central problem addressed in this thesis is therefore:

How can a complete, physically faithful, and reproducible software model of a small four-element UHF solar interferometer be constructed, and how can it be validated against both analytic theory and real archived solar radio data?

1.3 Objectives

The thesis pursues the following specific objectives:

1. **Antenna modelling.** Model the commercial UHF Yagi–Uda receiving element in 4NEC2 using the Method of Moments, extracting the input impedance, gain, and radiation pattern, and cross-validate the result against direct vector-network-analyser measurement and an independent finite-element solution. Quantify the mutual coupling between elements as a function of baseline and establish that it is negligible at the operating separations.
2. **End-to-end signal pipeline.** Implement a Python pipeline that converts real SWS Learmonth 610 MHz flux into complex antenna voltages using the Friis reception relation and Johnson–Nyquist noise, applies per-antenna geometric phase, cross-correlates to form visibilities, and inverts them to a dirty image.
3. **General array geometry.** Replace any one-dimensional, east–west-only assumptions with a general local-to-equatorial $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation so that arbitrary two-dimensional layouts can be simulated.
4. **Validation.** Verify that the simulated instrument reproduces the expected analytic signatures—the fringe period, the jinc-shaped visibility of a resolved disk, and the dirty image as the sky convolved with the synthesised beam—for the transit of 28 April 2024 at Learmonth.

1.4 Contributions

The principal contributions of this thesis are:

1. A 4NEC2 Method-of-Moments model of a low-cost commercial UHF Yagi–Uda element, cross-validated against direct measurement and an independent finite-element solution, together with a quantitative mutual-coupling study showing that the elements are effectively decoupled across the operational baselines—the coupling falling below -78 dB beyond 5 m, so that the array may be treated as a set of independent sensors.
2. A modular, reproducible interferometry simulation engine that is correct for *arbitrary* two-dimensional array geometries through a single, consistent (u, v, w) formulation applied identically to per-antenna phase, baseline computation, and fringe stopping.
3. A demonstration that the simulated instrument, driven by real archived solar flux, reproduces the classical analytic results of aperture synthesis, thereby validating both the instrument concept and the simulation itself as a design tool.

1.5 Organization of the Thesis

The remainder of this thesis is organised into five chapters. **Chapter 2** reviews the theoretical background: the radio emission of the Sun, the fundamentals of antenna theory, receiver noise, the principle of interferometry and the van Cittert–Zernike theorem, and aperture-synthesis imaging. **Chapter 3** presents the electromagnetic modelling of the Yagi–Uda receiving element and the mutual-coupling study of the array. **Chapter 4** describes the design of the end-to-end simulation pipeline, from flux-data acquisition through voltage synthesis, array geometry, correlation, and imaging. **Chapter 5** reports and analyses the simulation results and the validation against analytic predictions and the 28 April 2024 transit. The closing chapter, **Conclusion and Future Work**, summarises the findings and outlines possible extensions. References and appendices follow.

CHAPTER 2. THEORETICAL BACKGROUND

This chapter establishes the physical and mathematical foundations on which the rest of the thesis is built. It is organised around five themes: the radio emission of the Sun (Section 2.1); the fundamentals of antenna theory relevant to the receiving element (Section 2.2); the noise behaviour of a radio-astronomical receiver (Section 2.3); the principle of interferometry, developed from first principles starting with the mutual coherence function and the two-element correlator and proceeding to the visibility function, the (u, v, w) coordinate system, and the van Cittert–Zernike theorem (Section 2.4); and aperture-synthesis imaging, covering uv -coverage, Earth-rotation synthesis, and the dirty-image/dirty-beam formalism (Section 2.5). Section 2.6 summarises the practical error sources that distinguish a real instrument from its idealised description. The treatment of interferometry and synthesis imaging follows closely the formalism of Taylor et al. [1], *Synthesis Imaging in Radio Astronomy II*, whose Chapters 1–3 are the principal reference for Sections 2.4 and 2.5. Equations are numbered within the chapter for cross-reference in the later design and results chapters.

2.1 Solar Radio Emission

At radio wavelengths the Sun is not the sharp-edged disk seen in visible light but a stratified, optically thick plasma whose apparent size and brightness depend strongly on the observing frequency [2, 3]. Three atmospheric layers matter to a radio observer: the photosphere (the 5800 K optical surface, dominant only at millimetre wavelengths), the chromosphere, and the hot, tenuous corona, whose electron temperature reaches $\sim 10^6$ K. At metre and decimetre wavelengths the radiation we receive originates predominantly in the corona and upper chromosphere.

2.1.1 The Quiet Sun and the Brightness Temperature

In the absence of activity, the *quiet Sun* radiates by thermal free–free (bremsstrahlung) emission. Its intensity is conveniently expressed as a *brightness temperature* T_b , defined through the Rayleigh–Jeans law as the temperature of a blackbody that would produce the observed specific intensity I_ν at frequency ν :

$$I_\nu = \frac{2k_B\nu^2}{c^2} T_b, \quad (2.1)$$

where k_B is Boltzmann’s constant and c the speed of light. At 610 MHz the quiet-Sun brightness temperature is of order 10^5 – 10^6 K, and the integrated flux density is a few tens of solar flux units, where one solar flux unit is

$$1 \text{ sfu} = 10^{-22} \text{ W}/(\text{m}^2 \text{ Hz}). \quad (2.2)$$

This quiet, slowly varying background is the signal used for instrument validation in this thesis: its stability is an advantage, because departures from the predicted fringe pattern can then be

attributed to the instrument rather than to the source.

2.1.2 Solar Radio Bursts

Superimposed on the quiet background are transient *solar radio bursts*, which can raise the flux by several orders of magnitude. Following the classical morphological scheme they are grouped into five types (I–V) according to their appearance in the dynamic spectrum [4, 5]: long-lived Type I noise storms; slowly drifting Type II bursts associated with shock waves; fast-drifting Type III bursts from electron beams; broadband Type IV continua following flares; and short Type V continua. At decimetre wavelengths the emission is frequently *non-thermal*, produced by plasma or gyrosynchrotron mechanisms, and is spatially compact”which is precisely what makes a burst a meaningful target for a small interferometer.

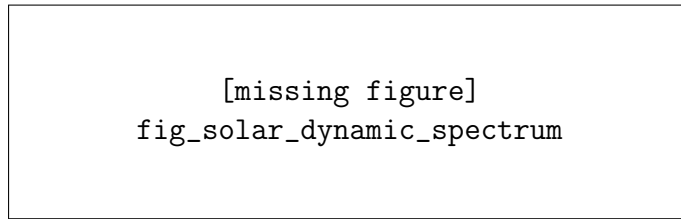


Figure 2.1. Schematic dynamic spectrum of the Sun showing the quiet-Sun background and the classification of bursts (Types I–V) in the frequency–time plane. The 610 MHz observing channel of this thesis is indicated.

2.2 Fundamentals of Antenna Theory

2.2.1 The Antenna as an Electromagnetic Transducer

A receiving antenna converts the incident electromagnetic field into a voltage at its terminals. Its directional sensitivity is described by the *normalised power pattern* $P_n(\theta, \phi)$, which for a receiving element is identical to its transmitting pattern by reciprocity. The *directivity* D and *gain* G follow from

$$D = \frac{4\pi}{\iint_{4\pi} P_n(\theta, \phi) d\Omega}, \quad G = \eta D, \quad (2.3)$$

where η is the radiation efficiency. The *half-power beamwidth* (HPBW) is the angular width over which $P_n \geq 0.5$, and it sets the field of view of a single element: in an interferometer the element pattern acts as the *primary beam* that multiplies the sky brightness.

2.2.2 Effective Aperture and the Friis Reception Relation

For imaging purposes the key quantity is the *effective aperture* $A_e(\theta, \phi)$, related to the gain by

$$A_e = \frac{\lambda^2}{4\pi} G. \quad (2.4)$$

If a source of flux density S_ν (in $\text{W}/(\text{m}^2 \text{Hz})$) is observed over a bandwidth $\Delta\nu$, the power delivered to a matched load is, by the Friis reception relation,

$$P_{\text{rx}} = \frac{1}{2} A_e S_\nu \Delta\nu, \quad (2.5)$$

the factor $\frac{1}{2}$ accounting for a single received polarisation of unpolarised radiation. Equation (2.5) is the bridge between the astronomical flux scale and the receiver, and it is the relation used in Chapter 4 to convert the SWS flux record into antenna voltages.

2.2.3 The Yagi–Uda Antenna

The receiving element used throughout this thesis is a Yagi–Uda array: a single driven element (here a folded dipole), a reflector behind it, and several directors in front. The parasitic elements re-radiate with phases set by their length and spacing, producing an end-fire pattern with gain typically in the range 8 dBi to 12 dBi. The folded dipole raises the driven-element impedance to roughly four times that of a simple dipole, which eases matching to the feed. The relevant figures of merit for our element are its input impedance Z_{in} , its gain and HPBW, and its standing-wave ratio

$$\text{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad \Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}, \quad (2.6)$$

where Z_0 is the feed-line impedance (75Ω for the coaxial cable used). These quantities are computed for the real element in Chapter 3.

2.3 Receiver Noise and Sensitivity

2.3.1 Johnson–Nyquist Thermal Noise

Any resistive element at physical temperature T generates a fluctuating voltage. The available noise power in a bandwidth $\Delta\nu$ is, in the Rayleigh–Jeans limit,

$$P_n = k_B T \Delta\nu, \quad (2.7)$$

which underlies the convention of expressing every noise contribution as an equivalent *noise temperature*. The total *system temperature* T_{sys} sums the sky, ground, antenna, and receiver contributions, and the system-equivalent flux density follows from Eq. (2.5) as $\text{SEFD} = 2k_B T_{\text{sys}}/A_e$.

2.3.2 The Radiometer Equation

The fundamental sensitivity limit of a total-power radiometer is the *radiometer equation*, which gives the smallest detectable change in antenna temperature after integrating for a time τ over a bandwidth $\Delta\nu$:

$$\Delta T_{\text{rms}} = \frac{T_{\text{sys}}}{\sqrt{\Delta\nu \tau}}. \quad (2.8)$$

The $1/\sqrt{\Delta\nu\tau}$ scaling—the improvement of signal-to-noise with the square root of the number of independent samples—is the statistical foundation on which both single-dish detection and interferometric correlation rest, and it is reproduced explicitly by the noise model of Section 4.3.

2.4 Radio Interferometry

The theory of radio interferometry is, at its core, the theory of measuring the *spatial coherence* of an electromagnetic field generated by an incoherent astronomical source. This section follows the development given in Chapters 1–3 of Taylor et al. [1], building the formalism from the mutual coherence function of the radiation field, through the operation of a two-element correlator, to the visibility function and the van Cittert–Zernike theorem that underlies all of aperture-synthesis imaging.

2.4.1 The Mutual Coherence Function of an Incoherent Source

Consider a monochromatic, scalar electromagnetic field $E(\mathbf{r}, t)$ radiated by a distant astronomical source and observed at two points $\mathbf{r}_1$ and $\mathbf{r}_2$ in the far field. The *mutual coherence function* of the field between these two points is defined as the time-averaged cross-correlation

$$V_{12}(\tau) \equiv \langle E(\mathbf{r}_1, t) E^*(\mathbf{r}_2, t - \tau) \rangle, \quad (2.9)$$

where the angle brackets denote a time average over an interval long compared with the inverse bandwidth of the radiation but short compared with any timescale of source variability, and τ is a relative time delay introduced between the two signal paths. A celestial source is composed of a vast number of independent radiating regions—electrons gyrating incoherently in the corona, for instance—so the radiation received from any two distinct directions on the sky is statistically independent. This property, that the source is *spatially incoherent*, is the single most important physical assumption in interferometry: it implies that the coherence measured *between two antennas* does not arise because the source itself is coherent, but because the same incoherent field is sampled at two separated points and the geometric delay between those points encodes information about the source’s angular structure. This is precisely the principle exploited eight decades ago by the Michelson stellar interferometer, in which two mirror apertures, fed into a common eyepiece, produce fringes whose contrast (the *degree of coherence*) falls as the angular size of the source increases relative to λ/b —and it is the optical-astronomy ancestor of the radio-interferometric technique developed in this thesis.

For two antennas separated by a baseline vector $\mathbf{b} = \mathbf{r}_1 - \mathbf{r}_2$, the mutual coherence function depends, for a quasi-monochromatic field of centre frequency ν and a source confined to a small range of directions $\hat{\mathbf{s}}$ about a phase centre, almost entirely on the projection of $\mathbf{b}$ along $\hat{\mathbf{s}}$. Writing $V_{12}(\tau)$ in terms of the source’s angular brightness distribution $I(\hat{\mathbf{s}})$ integrated over the

sky Ω gives the most general statement of the *coherence relation*:

$$V_{12}(\tau) = \int_{\Omega} I(\hat{\mathbf{s}}) \exp\left[-i 2\pi\nu\left(\tau - \frac{\mathbf{b} \cdot \hat{\mathbf{s}}}{c}\right)\right] d\Omega. \quad (2.10)$$

Equation (2.10) already contains the essential physics of synthesis imaging: the coherence function measured on a baseline $\mathbf{b}$ is a weighted, phase-shifted integral of the sky brightness, and once a slowly varying envelope and exact geometric compensation are introduced below it reduces to the Fourier-transform relation that all of imaging is built upon.

2.4.2 The Two-Element Interferometer and the Correlator

The simplest interferometer consists of two identical antennas, separated by a baseline vector $\mathbf{b}$, whose voltage outputs are multiplied and time-averaged by a device called a *correlator*. Consider first a single point source in direction $\hat{\mathbf{s}}$. The wavefront from the source reaches one antenna later than the other by the *geometric delay*

$$\tau_g = \frac{\mathbf{b} \cdot \hat{\mathbf{s}}}{c}. \quad (2.11)$$

In a real instrument this delay is compensated electronically—a process called *fringe stopping*—by inserting a compensating delay τ_g in one signal path before correlation, so that the residual delay tracks only the difference between the true geometric delay and the model delay used by the correlator. If the two voltage signals $V_1(t)$ and $V_2(t)$ from a quasi-monochromatic source of centre frequency ν_0 are each written as a slowly varying complex envelope multiplying the carrier, the correlator output for a point source, after fringe stopping and averaging over many carrier cycles, is the real part of a complex response:

$$r(\tau_g) \propto \cos(2\pi\nu_0 \tau_g) = \cos\left(\frac{2\pi}{\lambda} \mathbf{b} \cdot \hat{\mathbf{s}}\right). \quad (2.12)$$

This is the celebrated *interference fringe*: as the Earth rotates, $\hat{\mathbf{s}}$ sweeps slowly across the sky relative to $\mathbf{b}$, and the correlator output traces out a slowly varying cosine whose instantaneous frequency is the *fringe rate*. In practice a *complex correlator* forms both an in-phase output (cos) and a quadrature output (sin, obtained by inserting a quarter-wave phase shift in one path before the second multiplication), so that the full complex response—amplitude and phase—of the source on that baseline is recovered, rather than only its real part. This complex output is precisely the visibility introduced in Section 2.4.4 below.

For an east–west baseline of length b observing a source at hour angle H and declination δ , geometric differentiation of Eq. (2.11) with respect to time gives the rate of change of fringe phase, and hence the fringe period

$$T_{\text{fringe}} \approx \frac{\lambda}{b \omega_{\oplus} \cos \delta}, \quad (2.13)$$

where $\omega_{\oplus}$ is the Earth’s sidereal rotation rate. For $b = 15$ m, $\lambda = 0.49$ m and the solar dec-

lination of late April, this evaluates to roughly eight minutesâ€”a prediction tested directly in Chapter 5. The fringe period is the most basic and most directly verifiable analytic signature of an interferometer, and for that reason it is the first benchmark used to validate the simulation.

2.4.3 Quasi-Monochromatic Approximation and the Effect of Bandwidth

The derivation above implicitly assumed a single frequency ν_0 . In practice every receiver has finite bandwidth $\Delta\nu$, and Eq. (2.10) must be integrated over the receiver passband. If the residual delay error after fringe stopping is $\Delta\tau = \tau_g - \tau_{\text{model}}$, integrating a uniform passband of width $\Delta\nu$ centred on ν_0 introduces a multiplicative attenuation factor of the fringe amplitude,

$$\text{sinc}(\Delta\nu \Delta\tau) = \frac{\sin(\pi \Delta\nu \Delta\tau)}{\pi \Delta\nu \Delta\tau}, \quad (2.14)$$

known as *bandwidth smearing* or *delay-beam* attenuation. It reduces the sensitivity of the interferometer to sources far from the phase-tracking centre, in direct analogy to the way a finite integration time attenuates response to sources far from the field centre once Earth rotation is taken into account (*time-average smearing*). Because the residual delay error in this simulation is determined entirely by the known geometry rather than by an imperfect delay model, both effects are small for the narrow field of the quiet Sun and are quantified in Section 2.6 and revisited in Chapter 4.

2.4.4 The Visibility Function

For an extended, spatially incoherent source, the quantity actually measured by the complex correlatorâ€”the quasi-monochromatic, geometrically compensated form of the mutual coherence function of Eq. (2.10)â€”is the *complex visibility* V . Following Taylor et al. [1], it is conventional to measure the baseline components in wavelengths in a coordinate system (u, v, w) with u toward the east, v toward the north, and w toward the phase-tracking centre, and to write the source position in terms of direction cosines (l, m, n) relative to that centre, with $n = \sqrt{1 - l^2 - m^2}$. In this notation the visibility integral becomes

$$V(u, v, w) = \iint A_n(l, m) I(l, m) e^{-2\pi i [ul + vm + w(n-1)]} \frac{dl dm}{n}, \quad (2.15)$$

where $I(l, m)$ is the sky brightness distribution and $A_n(l, m)$ is the normalised power pattern of a single antenna (the *primary beam* of Section 2.2), which weights the contribution of each sky direction according to the antenna's directional sensitivity. Equation (2.15) is the central relation of interferometry: a single baseline at a single instant measures one complex number, $V(u, v, w)$, which is a weighted, projected Fourier component of the sky brightness. An array of N antennas observing simultaneously measures $N(N - 1)/2$ such numbers, one per baseline pair, and it is the accumulation of these samplesâ€”over many baselines and over time as the Earth rotatesâ€”that allows the sky brightness to be reconstructed.

2.4.5 The (u, v, w) Coordinate System

The baseline coordinates (u, v, w) in Eq. (2.15) are obtained from the physical antenna positions by a sequence of two coordinate transformations, as described by Taylor et al. [1]. First, the local antenna positions, measured in an East-North-Up (ENU) frame tangent to the Earth at the observing site, are converted to an Earth-fixed equatorial (X, Y, Z) frame, in which X points to the intersection of the equator and the local meridian, Y points east, and Z points to the celestial pole. Second, this equatorial frame is rotated, for the instantaneous hour angle H and declination δ of the phase-tracking centre, into the baseline-dependent (u, v, w) frame in which w points along the source direction:

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} \sin H & \cos H & 0 \\ -\sin \delta \cos H & \sin \delta \sin H & \cos \delta \\ \cos \delta \cos H & -\cos \delta \sin H & \sin \delta \end{bmatrix} \begin{bmatrix} X_\lambda \\ Y_\lambda \\ Z_\lambda \end{bmatrix}, \quad (2.16)$$

where the subscript λ indicates that the equatorial baseline components have been expressed in units of wavelength. As the hour angle H advances through the day because of Earth's rotation, a fixed physical baseline traces out a continuously varying (u, v, w) track—the basis of Earth-rotation synthesis discussed in Section 2.5. This single, general transformation is applied identically to every baseline and every instant in the simulation pipeline, so that per-antenna geometric phase, instantaneous baseline length, and fringe-stopping delay are all computed from one consistent geometric model, valid for arbitrary two-dimensional array layouts rather than only for the east–west case of Eq. (2.11).

2.4.6 The Van Cittert–Zernike Theorem and the Small-Field Approximation

The cornerstone of modern synthesis imaging, as formulated in Taylor et al. [1] (and ultimately a radio-astronomical restatement of the van Cittert–Zernike theorem of classical coherence theory), follows from Eq. (2.15) in the limit of a narrow field of view. If the extent of the primary beam $A_n(l, m)$ is small enough that $l, m \ll 1$ over the region where $A_n I \neq 0$, then $n - 1 = \sqrt{1 - l^2 - m^2} - 1 \approx -\frac{1}{2}(l^2 + m^2)$ is small, and provided the w -term phase error $2\pi w(n - 1)$ remains $\ll 1$ radian across the field of view, the w -dependence of Eq. (2.15) may be dropped entirely. The visibility integral then collapses to an exact two-dimensional Fourier-transform pair between the modified sky brightness $A_n I/n$ and the visibility:

$$V(u, v) = \iint A_n(l, m) I(l, m) e^{-2\pi i(ul+vm)} dl dm. \quad (2.17)$$

This is the form of the van Cittert–Zernike theorem used throughout synthesis imaging: the visibility and the (apodised) sky brightness are Fourier conjugates, so that, given sufficient and suitable sampling of $V(u, v)$, the sky brightness can in principle be recovered by inverse Fourier transformation. For the solar disk, whose $32'$ angular diameter subtends $l, m \sim 5 \times 10^{-3}$, and for the short, near-zenith baselines used in this thesis, the w -term is indeed negligible and

Eq. (2.17) applies without correction—a simplification that would not be valid for a wide-field or long-baseline array and is revisited quantitatively in Section 2.6.

Equation (2.17) shows that each baseline samples one Fourier (spatial-frequency) component of the apodised sky brightness, with (u, v) playing the role of spatial frequency conjugate to the angular coordinates (l, m) : a baseline of length $|\mathbf{b}|$ measured in wavelengths corresponds to an angular scale of order $1/|\mathbf{b}|_\lambda$ radians on the sky, exactly analogous to the resolution argument of Section 2.1. For a uniformly bright circular disk of angular radius θ_s (a simplistic but effective model for the quiet Sun), the two-dimensional Fourier transform in Eq. (2.17) has the closed analytic form

$$V(q) = \frac{2J_1(2\pi q\theta_s)}{2\pi q\theta_s} \equiv \text{jinc}(2\pi q\theta_s), \quad q = \sqrt{u^2 + v^2}, \quad (2.18)$$

where J_1 is the first-order Bessel function of the first kind. The jinc function falls monotonically from unity at $q = 0$ (the disk’s total flux, measured on a zero-length baseline) to its first null at $q\theta_s \approx 0.61$, beyond which the visibility amplitude is small and oscillatory. This closed form provides a direct, analytic benchmark against which the simulated visibilities of the resolved solar disk are validated in Chapter 5: a correctly implemented simulation must reproduce both the jinc envelope and the location of its first null as a function of baseline length.

2.5 Aperture Synthesis Imaging

2.5.1 uv -Coverage and Earth-Rotation Synthesis

The set of (u, v) points sampled by an interferometric array—its *uv-coverage*—dictates the array’s imaging capability, in direct analogy to the way the aperture of a filled-aperture telescope determines its point-spread function. A single snapshot with N antennas yields $N(N - 1)/2$ instantaneous baselines, hence $N(N - 1)/2$ independent samples of $V(u, v)$ (each baseline contributing two complex-conjugate samples at $\pm(u, v)$, since $I(l, m)$ is real and therefore $V(-u, -v) = V^*(u, v)$). For the four-element array of this thesis, $N = 4$ gives only six instantaneous baselines—far too sparse, on its own, to constrain a two-dimensional image.

As the Earth rotates, however, the projection of any fixed physical baseline onto the plane perpendicular to the source direction changes continuously, because the rotation matrix in Eq. (2.16) depends on the hour angle H . For a baseline with equatorial components $(X_\lambda, Y_\lambda, Z_\lambda)$ observing a source at fixed declination δ , eliminating H between the u and v equations of Eq. (2.16) shows that each baseline traces an *ellipse* in the (u, v) plane as H varies over the day, with semi-major axis equal to the projected east–west baseline length and semi-minor axis foreshortened by $\sin \delta$. This technique, known as *Earth-rotation synthesis* [1], allows a small number of physical antennas to synthesise, over the course of a transit, a uv -coverage equivalent to a much larger number of simultaneous baselines, without any moving parts. It is the reason a four-element, six-baseline array can nonetheless produce a meaningful synthesised image once data accumulated over an observing transit are combined—a trade-off explored quantitatively, for the compact east–west geometry of the physical array against more general two-dimensional configurations,

in Chapter 5.

2.5.2 The Sampling Function and the Convolution Equation

Because only a finite, discrete set of (u, v) points is ever sampled, it is convenient to describe the measurement formally as a multiplication of the true, continuous visibility function by a *sampling function* $S(u, v)$ —a sum of Dirac delta functions (optionally weighted to account for the relative sensitivity or density of different baselines) located at each measured (u, v) point and its conjugate. The measured visibility is then $S(u, v) V(u, v)$, and inverse-Fourier-transforming this sampled, rather than continuous, visibility does not yield the true sky brightness I , but rather the *dirty image* I_D :

$$I_D(l, m) = \iint S(u, v) V(u, v) e^{+2\pi i(ul+vm)} du dv. \quad (2.19)$$

By the convolution theorem, multiplication of V by S in the Fourier (uv) domain corresponds to convolution in the image domain, so Eq. (2.19) may equivalently, and more usefully, be written as the *Convolution Equation* of synthesis imaging [1]:

$$I_D(l, m) = I(l, m) * B(l, m), \quad (2.20)$$

where $*$ denotes two-dimensional convolution and the *dirty beam* (equivalently, the *point-spread function*, PSF, or *synthesised beam*) $B(l, m)$ is simply the Fourier transform of the sampling function itself,

$$B(l, m) = \iint S(u, v) e^{+2\pi i(ul+vm)} du dv. \quad (2.21)$$

Equation (2.20) expresses the fundamental imaging equation of an interferometer: the dirty image is exactly what one would expect from imaging the true sky through an instrument whose response to a point source is the beam $B(l, m)$ —a direct radio analogue of the diffraction-limited PSF of an optical telescope, except that here the PSF is determined not by a filled aperture but by the discrete pattern of sampled (u, v) points.

2.5.3 Properties of the Dirty Beam

Because $S(u, v)$ is a sparse, typically non-uniform set of samples rather than a filled, azimuthally symmetric disk (as for a filled-aperture telescope), the dirty beam $B(l, m)$ generally has a narrow central peak—whose width sets the nominal angular resolution of the synthesised image, of order $1/b_{\max}$ radians for a maximum baseline $b_{\max}$ measured in wavelengths—surrounded by sidelobes that reflect the gaps in uv -coverage. For the compact, six-baseline east–west array of this thesis, the uv -coverage accumulated over an Earth-rotation transit consists of six partial elliptical arcs rather than a densely filled disk, so $B(l, m)$ retains significant sidelobe structure; this is expected, and the resulting dirty image is still a rigorously correct image of the sky under Eq. (2.20) given the actual sampling achieved; it is simply not a clean, sidelobe-free representation of $I(l, m)$. Because the simulation in this thesis is observing a single, compact, time-stable

calibrator (the quiet-Sun disk) rather than attempting to image a complex field crowded with confusing sources, the dirty image and dirty beam together are sufficient to validate the instrument: demonstrating numerically that the simulated I_D equals the assumed input sky map convolved with the independently computed $B(l, m)$, as required by Eq. (2.20), constitutes the ultimate end-to-end mathematical verification of the entire signal-processing pipeline, from antenna voltages to final image, and is carried out explicitly in Chapter 5.

2.6 Practical Error Sources

A real interferometer departs from the idealised description above in several ways that the simulation must either model or justify neglecting:

- **Mutual coupling** between closely spaced elements distorts the primary beam and the impedance; it is quantified for the array in Chapter 3.
- **Bandwidth and time-average smearing** attenuate visibilities far from the phase centre; with fringe stopping and the narrow field of the Sun these are small but are estimated in Chapter 4.
- **Phase-calibration and pointing errors** shift and blur the image; in simulation the geometry is known exactly, so these are treated as controllable perturbations.
- **Ionospheric phase** is, for baselines $\lesssim 130$ m at 610 MHz, essentially common-mode and cancels in the differential measurement, with a residual differential phase of about one degree.

2.7 Chapter Conclusion

This chapter has assembled the theory required for the remainder of the thesis: the nature of solar radio emission and the choice of 610 MHz; the antenna parameters that define the receiving element and primary beam; the noise statistics that set the sensitivity; and, at greater length, the physics of interferometry built up from the mutual coherence function of an incoherent source (Eq. (2.10)), through the operation of the two-element correlator and the interference fringe (Eqs. (2.11)–(2.13)), to the visibility function and the general (u, v, w) transformation of Eq. (2.16), the van Cittert–Zernike relation of Eq. (2.17), and the analytic jinc visibility of a uniform disk (Eq. (2.18)). Building on this, the aperture-synthesis framework of Section 2.5 established how Earth rotation fills the uv -plane and how the resulting sampling function gives rise to the dirty image and dirty beam of Eqs. (2.20)–(2.21). This formalism, developed throughout in the notation of Taylor et al. [1], provides both the modelling choices and the analytic benchmarks—the fringe period, the jinc-shaped visibility decay, and the convolution relation between dirty image, true sky, and synthesised beam—used to validate the simulation in the chapters that follow.

CHAPTER 3. ELECTROMAGNETIC MODELLING OF THE RECEIVING ELEMENT

The fidelity of the entire interferometric simulation rests, before anything else, on a mathematically and physically sound model of the receiving element. This is not a peripheral concern: in a synthesis array the single-element radiation pattern *is* the primary beam of the instrument, fixing both the field of view within which the Sun can be observed and the angular envelope through which off-axis terrestrial interference leaks into the measurement. Equally, the electromagnetic coupling between neighbouring elements determines whether the array may be treated as a set of independent sensors—the premise on which the whole van Cittert–Zernike imaging framework of Chapter 2 is built—or whether it must be modelled as a single coupled network with a correspondingly more complex calibration.

This thesis is, at its core, a study of an *antenna system*: the solar radio burst is the observational target, but the scientific contribution lies in the rigorous characterisation of the receiving array and in demonstrating that its electromagnetic behaviour justifies the assumptions adopted downstream in the signal-processing pipeline. The present chapter therefore occupies a central position in the work. It proceeds as follows. Section 3.1 develops the theoretical background of the Yagi–Uda antenna at UHF, with emphasis on parasitic excitation, the folded dipole, and the reasons this antenna class is well matched to a low-cost 610 MHz solar instrument. Section 3.2 describes the physical hardware and the preliminary field deployment at Xuan Mai, and uses the field measurements to draw a sharp distinction between the receiving *element* (which performed correctly) and the receiving *system* (which did not); this distinction is the motivation for the modelling effort that follows. Section 3.3 establishes why mutual-coupling analysis is not optional but mandatory for an aperture-synthesis array. Section 3.4 then presents the complete 4NEC2 methodology: the Method-of-Moments formulation, the construction of the wire model, the extracted single-element parameters, a three-way cross-validation against an independent finite-element solution and against direct measurement, the mutual-coupling sweep across operational baselines, and finally the extraction of the primary beam into the numerical format consumed by the imaging pipeline.

3.1 Theoretical Background of the Yagi–Uda Antenna

The Yagi–Uda antenna remains one of the most widely used directional antennas in the Ultra-High-Frequency (UHF) band. First described in the English-language literature by Yagi [6], building on the work of Shintaro Uda, the design achieves an attractive balance between high directivity, mechanical robustness, and very low fabrication cost [7, 8]. These properties are precisely what a low-budget, educational solar radio instrument requires, and they are the reason a commercial Yagi forms the receiving element of this project.

3.1.1 Operating Principle and Parasitic Excitation

Unlike a phased array, in which every element is fed with an individually controlled voltage and phase, the Yagi–Uda array operates by *parasitic excitation*. It comprises a single active *driven element* together with several passive (parasitic) elements whose currents are induced purely by electromagnetic coupling.

The interaction is governed by the mutual-impedance description of coupled antennas [7]. For an array of $N + 1$ elements, indexing the reflector as -1 , the driven element as 0 , and the directors as $1, 2, \dots, N$, the terminal voltages and currents satisfy a linear system,

$$V_m = \sum_n Z_{mn} I_n, \quad (3.1)$$

where I_n is the current on element n , Z_{mn} ($m \neq n$) is the mutual impedance between elements m and n , and Z_{mm} is the self-impedance. Only the driven element is fed, so $V_0 \neq 0$ while every parasitic element is short-circuited at its (virtual) terminals:

$$0 = \sum_n Z_{kn} I_n \quad (k \neq 0). \quad (3.2)$$

Equation (3.2) shows that the parasitic currents are not free parameters: they are fixed by the geometry through the off-diagonal mutual impedances. The design freedom available to the engineer is therefore the choice of element lengths and spacings, which set the magnitude and—crucially—the *phase* of each induced current.

The directional behaviour follows directly from these induced phases [8, 9]:

- **Reflector.** Positioned behind the driven element and made slightly longer than the resonant half-wavelength ($\approx 5\%$ longer), the reflector presents an *inductive* reactance. The induced current consequently lags the driving field, and the superposition of the re-radiated field with the primary field is constructive toward the front and destructive toward the rear. A single reflector captures most of the available rearward suppression; additional reflectors yield diminishing returns, which is why practical Yagis use one reflector (or, as here, one reflecting structure) but many directors.
- **Directors.** Positioned in front of the driven element and made $\approx 5\%$ *shorter* than resonance, the directors are *capacitive*. Their induced currents lead the driving field, so the composite wave is progressively guided forward, sharpening the main lobe into an end-fire beam along the boom axis [7].

The net effect is a travelling-wave-like structure that concentrates radiated (or, by reciprocity, received) power along the boom.

3.1.2 Directivity Build-Up and the Gain–Bandwidth Trade-off

The directivity of a Yagi–Uda array increases monotonically, though sub-linearly, with the number of directors. A useful qualitative progression, consistent with both classical treatments and

the present design, is that an isolated half-wave dipole provides about 2.15 dBi; adding a reflector raises this to roughly 7 dBi; the first director takes it close to 9.7 dBi; and a well-tuned six-director structure of the type used here reaches the 9.5 dBi to 13 dBi range depending on boom length and tuning [7, 9]. Each added director narrows the main lobe, which improves angular resolution and—important for a ground-based solar instrument—reduces the half-power beamwidth through which horizon-level terrestrial interference can enter.

This gain, however, is purchased at the expense of bandwidth and of tuning sensitivity. As more directors are added, the array impedance becomes increasingly reactive and the input match degrades rapidly away from the design frequency; aggressive gain optimisation can reduce usable bandwidth and lower the peak gain by several decibels if the element phasing drifts from the optimum [9]. For the solar application this trade-off is benign: the instantaneous observing bandwidth is narrow (a single 1 MHz channel centred at 610 MHz), so a moderately high-gain, moderately broadband design is entirely appropriate, and the bandwidth inherited from the original DVB-T2 service is far wider than the observation requires.

3.1.3 The Folded Dipole Driven Element

A standard half-wave dipole presents a radiation resistance near $73\ \Omega$ in isolation, but this value drops substantially once the element is loaded by closely spaced parasitic directors, often falling well below $30\ \Omega$. Such a low, geometry-sensitive impedance is difficult to match robustly over a useful band. The receiving element of this project therefore uses a *folded dipole* as its driven element, a choice that is standard in commercial UHF television antennas for exactly this reason [8].

A folded dipole consists of two closely spaced parallel conductors joined at both ends, with the feed breaking only one conductor at its centre. Because the structure carries both a transmission-line (even) mode and an antenna (odd) mode, and the radiating current divides between the two arms, the current drawn at the feed terminals for a given radiated power is halved relative to a simple dipole. The input impedance is consequently stepped up by a factor of approximately four [7, 8]:

$$Z_f \approx 4 Z_d \approx 4 \times 73\ \Omega \approx 292\ \Omega. \quad (3.3)$$

Two practical advantages follow. First, the resulting impedance ($\sim 300\ \Omega$) is high enough that the parasitic loading of the directors brings it down into a convenient range rather than collapsing it toward zero, giving a more stable, broadband match. Second, the $\sim 300\ \Omega$ balanced impedance is transformed to the standard $75\ \Omega$ unbalanced coaxial feed by a simple 4:1 balun, which simultaneously suppresses common-mode currents on the outer conductor of the cable. The combination of a folded dipole, a 4:1 balun, and a $75\ \Omega$ feed is the canonical UHF-television front end, and it is inherited unchanged by the present instrument.

3.1.4 Suitability for 610 MHz Solar Observation

The choice of 610 MHz as the observing frequency is driven by solar physics: metre-to-decimetre radio emission at this frequency originates in the low-to-middle corona, and 610 MHz is one of the discrete frequencies at which long-running solar patrol observatories, including the Learmonth station whose archived flux data are used for validation in this thesis, monitor solar radio bursts [4, 5]. At this frequency the free-space wavelength is

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8 \text{ m/s}}{610 \times 10^6 \text{ Hz}} \approx 491.5 \text{ mm}, \quad (3.4)$$

so a multi-element Yagi remains physically compact (boom length of order one metre) and can be hand-built and tripod-mounted—a decisive advantage for a low-cost, portable instrument.

The Sun is a weak source at these frequencies during quiet conditions, and the target Type II/Type III bursts must be separated from a strong terrestrial noise background. A directional element with moderate gain (~ 9.5 dBi) and a high front-to-back ratio is therefore valuable: it concentrates sensitivity toward the solar transit path while suppressing pickup from the horizon and from behind the array [10].

There is, however, an important double-edged aspect to using a commercial DVB-T2 antenna for this purpose, and it is worth stating explicitly because it shapes the field results in Section 3.2. The very features that make the antenna attractive—that it is mass-produced, inexpensive, and resonant across the 470 MHz to 862 MHz UHF television band—also make it maximally sensitive to terrestrial digital-television broadcasts that occupy exactly that band. In a populated area the 610 MHz observing channel sits in the middle of an active broadcast allocation, so the antenna efficiently collects precisely the interference one most wishes to avoid. This tension between availability and radio-frequency cleanliness is a recurring theme of the field campaign and one of the principal motivations for retreating to a controlled numerical model.

3.2 Real-World Antenna and Preliminary Hardware Setup

3.2.1 Antenna Specification (HKD G11KD-T2)

The physical receiving element adopted for this project is the commercial HKD G11KD-T2, a high-gain UHF Yagi–Uda antenna originally marketed for DVB-T2 digital-television reception. As a commercial-off-the-shelf (COTS) component it combines mature RF engineering with rugged, weather-resistant aluminium construction, and it is available at a small fraction of the cost of a purpose-built radio-astronomy feed.

Because the manufacturer publishes only marketing specifications, the geometry used for simulation was established by reconciling three independent sources: the manufacturer drawing, photographs of the deployed unit, and direct physical measurement of the antenna with callipers. The resulting model, summarised in Table 3.1, comprises an eleven-element structure on an aluminium boom of overall length ≈ 550 mm.

Table 3.1. Measured geometry of the HKD G11KD-T2 receiving element used as the basis for the electromagnetic model. Spacings are along the boom (director 1 measured from the driven element).

Element	Dimension
Reflector (four-bar U-frame)	outer bars at $z = \pm 40$ mm, inner bars at $z = \pm 10$ mm
Driven element (folded dipole)	length ≈ 210 mm, fold height 44 mm
Director lengths	≈ 210 mm (each)
Director spacings	82, 40, 50, 50, 66, 69 mm
Boom (square profile)	15×15 mm, length ≈ 550 mm
Element material	aluminium (hollow tube / solid rod)

The four-bar U-frame reflector is the distinguishing geometric feature of this antenna. Rather than a single straight rod, the reflector forms a shallow three-dimensional frame: this widens the effective reflecting aperture, raises the front-to-back ratio, and in the radio-astronomy context suppresses pickup of warm-ground thermal noise entering through the rear lobe.

3.2.2 Preliminary Field Deployment at Xuan Mai

To assess the feasibility of solar radio observation with this hardware, a preliminary two-element interferometer was deployed at the Citrus Research and Development Centre in Xuan Mai, Chuong My, Hanoi, over several sessions in December 2025. Two identical HKD G11KD-T2 antennas were mounted on heavy-duty photographic tripods, allowing manual adjustment in azimuth and elevation to follow the solar transit, and separated along an approximately east–west baseline. The radio-frequency chain comprised a mast-head Low-Noise Amplifier (LNA) at each antenna, powered through a custom dual-channel bias-tee that injected DC up the coaxial feed while passing the received 610 MHz signal to the receiver; the digitised back end used an RTL-SDR dongle controlled by a single-board computer, configured as a total-power radiometer.

The campaign confirmed that the instrument concept was sound in principle—the data showed a broad diurnal rise consistent with the Sun transiting the antenna beams—but it also exposed a series of operational failures that prevented any reliable detection of a genuine solar radio burst.

3.2.3 Element Versus System: Diagnosing the Field Failure

The most important outcome of the field campaign, for the purposes of this thesis, is a clean separation between the performance of the receiving *element* and the performance of the receiving *system*. The element was not at fault; the system was. The evidence is as follows.

The element is sound. During troubleshooting, the antennas and their LNAs were characterised directly at the feed point with a vector network analyser (NanoVNA). Each element returned a return loss $S_{11} \approx -12.8$ dB and a voltage standing-wave ratio VSWR ≈ 1.59 at

610 MHz, and the LNAs delivered their rated gain. These figures indicate a well-matched, correctly functioning receiving element. The radiating hardware, in other words, met its specification.

The system was degraded at the physical layer. The failures lay in the chain between the antenna terminals and the digitiser:

1. **Cable insertion loss.** Transmission-coefficient (S_{21}) measurements of the coaxial runs alone showed -3.1 dB to -3.5 dB of loss. More than half of the received power was therefore dissipated in the feed before reaching the receiver—fatal for a source already close to the noise floor.
2. **Bias-tee/LNA fault.** One mast-head LNA lost power through a disconnection in its bias-tee channel, leaving that arm of the interferometer effectively dead during part of the campaign.
3. **Terrestrial RFI and ground noise.** The outdoor noise floor was densely populated with carriers from nearby electronic equipment and broadcast services. As anticipated in Section 3.1.4, the DVB-T2 antenna’s in-band sensitivity made this interference unavoidable, and the recorded dynamic spectra showed strong, non-solar fringe-like structure and night-time oscillations driven by local sources.
4. **Short, ill-defined baseline and weak source.** The achievable physical baseline (~ 8 m in the field) and the faintness of the quiet-Sun signal (~ 0.5 dBm excursions) left little margin against the degradations above.

Implication. Because the element is provably sound while the system is degraded by interference, feed loss, and—as analysed in Chapter 5—the absence of phase coherence between two independent receiver chains, the correct engineering response is not to keep re-fabricating the field array by trial and error. It is to isolate the one component that can be characterised exactly, the antenna, in a controlled numerical environment where the geometry is known to the millimetre and where cabling, interference, and fabrication tolerance are absent. A rigorous electromagnetic model then provides the *ground truth* primary beam and the coupling figures that the field hardware could never supply directly, and it furnishes the clean, idealised receiving model on which the rest of the simulation depends. The remainder of this chapter develops that model.

3.3 The Rationale for Mutual-Coupling Analysis

3.3.1 The Independent-Sensor Assumption in Aperture Synthesis

The imaging framework adopted in this thesis—the van Cittert–Zernike theorem and the dirty-image/CLEAN machinery of Chapter 2—relies on a specific assumption: that each receiving element samples the spatial coherence function of the sky *independently*, with a primary beam

$A_n(l, m)$ identical to the isolated single-element pattern [10]. When this holds, the measured cross-correlation between two elements is a clean sample of the complex visibility, and the recovered phase is purely the geometric delay between the two apertures.

This assumption is not automatically valid. When antennas are placed in proximity, the field radiated and scattered by one element induces currents on the other—mutual coupling. For two identical elements the driving-point impedance of element 1 is modified by the presence of element 2,

$$Z_{1,\text{in}} = Z_{11} + Z_{12} \frac{I_2}{I_1}, \quad (3.5)$$

where Z_{11} is the self-impedance, Z_{12} the mutual impedance, and I_1, I_2 the terminal currents. The quantity on the left is the *active* (or scan) impedance—the impedance an element actually presents when the whole array is energised—and its departure from the isolated value Z_{11} is a direct, quantitative fingerprint of the coupling [7, 11].

3.3.2 Why Coupling Specifically Corrupts Interferometric Phase

For a total-power instrument, the chief consequence of coupling would be a mismatch loss. For an *interferometer* the consequence is more insidious, because the measurement is fundamentally a measurement of phase. Coupling perturbs the active impedance (mismatch and standing waves), distorts and makes asymmetric the primary beam, and—most damagingly—adds a parasitic, geometry-dependent phase term to the received signal. The interferometer requires that the measured phase be the pure geometric delay $\phi = \frac{2\pi}{\lambda} \mathbf{b} \cdot \hat{\mathbf{s}}$ between the elements; any additional coupling-induced phase contaminates the visibility and propagates directly into the dirty image as spurious structure [10, 11].

The standard ways to quantify coupling are equivalent and well established [7, 12]: the off-diagonal mutual impedance Z_{12} ; the transmission scattering parameter S_{21} (the same information expressed in incident/reflected waves, and the quantity most directly measurable with a network analyser); the full $N \times N$ coupling matrix for an N -element array; and the shift of the active impedance relative to the isolated element. It is therefore not sufficient to *assume* the elements are independent at the operating baseline—the assumption must be *demonstrated* by computing one of these measures as a function of element separation. That demonstration is the purpose of Section 3.4.5.

3.4 4NEC2 Simulation Methodology and Results

To obtain precise electromagnetic parameters free of the field’s confounding variables, the U-frame folded-dipole Yagi was modelled in 4NEC2, a widely used implementation of the Numerical Electromagnetics Code [13].

3.4.1 The Method of Moments in 4NEC2

4NEC2 solves the thin-wire Electric-Field Integral Equation (EFIE) by the Method of Moments (MoM) [13, 14]. The continuous antenna is discretised into short, straight wire segments; the unknown current distribution is expanded in basis functions on these segments; and enforcement of the boundary condition that the total tangential electric field vanish on the conducting surface converts the integral equation into a dense linear system $[Z][I] = [V]$, where $[Z]$ is the generalised impedance matrix, $[V]$ the excitation, and $[I]$ the segment currents to be solved. Accurate results require each segment to be short compared with the wavelength—typically between $\lambda/10$ and $\lambda/20$ —and the segment radius to satisfy the thin-wire approximation. Once the current distribution is known, the input impedance follows from the feed-segment voltage and current, and the far-field pattern from the radiation integral over all segments. MoM is exceptionally well suited to open wire structures such as a Yagi–Uda array: it discretises only the conductors (not the surrounding volume), converges rapidly, and yields impedance and pattern data of high accuracy in the UHF band.

3.4.2 Wire Model Construction

The HKD geometry of Table 3.1 was translated into a fourteen-wire NEC model: the four U-frame reflector bars, the folded-dipole driven element (modelled as its constituent wires), and the six directors, with conductor radii in the 2.0 mm to 3.5 mm range to match the physical tubing. The model was solved in free space (ground flag GN -1); this is the physically correct choice for a sky-pointing radio-astronomy antenna, whose intrinsic pattern is the quantity required, with the effect of the real ground treated separately rather than baked into the element pattern. The excitation was a 1 V source placed at the centre of the folded dipole (the central segment of the driven-element wire), i.e. at the physical feed point.

The impedance reference was set to $75\ \Omega$ throughout. This is not an arbitrary normalisation: the antenna is a $75\ \Omega$ DVB-T2 device, the mast-head LNA is a $75\ \Omega$ part, and the entire feed is $75\ \Omega$ coaxial cable, so $75\ \Omega$ is the impedance against which return loss and SWR are physically meaningful for this system.

3.4.3 Single-Element Results

Solving the model at 610 MHz yields the single-element parameters collected in Table 3.2.

Table 3.2. Simulated single-element parameters of the eleven-element U-frame Yagi at 610 MHz (4NEC2, free space).

Parameter	Value
Operating frequency	610 MHz ($\lambda \approx 491.5$ mm)
Input impedance Z_{in}	$108.4 - j22.0 \Omega$
Reflection coeff. $ \Gamma $ (at 75Ω)	≈ 0.215 (≈ -13.3 dB)
SWR (at 75Ω)	1.55
Forward gain	9.48 dBi
Half-power beamwidth (E-plane)	$\approx 52^\circ$
Radiation efficiency	$\approx 100\%$ (lossless, free space)

The input impedance is $Z_{\text{in}} = 108.4 - j22.0 \Omega$: a real part close to the 75Ω reference and a small capacitive reactance indicating the antenna is operating just below its exact resonance. Referenced to 75Ω , the reflection-coefficient magnitude is

$$|\Gamma| = \left| \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \right| = \left| \frac{(108.4 - j22.0) - 75}{(108.4 - j22.0) + 75} \right| \approx 0.215, \quad (3.6)$$

equivalent to a return loss of about -13.3 dB, and the standing-wave ratio follows from Equation (2.6) as

$$\text{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \approx 1.55. \quad (3.7)$$

An SWR of 1.55 (comfortably below the conventional 2.0 acceptability threshold) means that the great majority of incident power is delivered into the feed with negligible reflective loss, and that no additional matching network is required—the folded-dipole/4:1-balun front end matches the array to 75Ω on its own.

The antenna delivers a forward gain of 9.48 dBi, nearly an order of magnitude above an isotropic radiator, with an E-plane half-power beamwidth of about 52° (Figure 3.1). This beamwidth is a good compromise for an un-tracked or hand-tracked solar instrument: narrow enough to reject horizon noise and to give the U-frame reflector its high front-to-back ratio, yet wide enough that the Sun dwells within the main lobe for a useful fraction of the day without continuous mechanical steering.



Figure 3.1. Simulated E-plane radiation pattern of the receiving element (4NEC2, free space), showing the end-fire main lobe, the high front-to-back ratio provided by the U-frame reflector, and the $\sim 52^\circ$ half-power beamwidth that defines the interferometer primary beam.

3.4.4 Cross-Validation: Measurement, MoM, and FEM

A simulated result carries weight only if it is corroborated. A particular strength of this study is that the single-element match is confirmed by *three independent methods* resting on entirely different principles: a direct physical measurement, the Method of Moments, and an independent finite-element solution obtained by a partner hardware group on the same antenna. Table 3.3 collects the three.

Table 3.3. Three-way cross-validation of the single-element match at 610 MHz: physical measurement, Method of Moments (this work), and an independent finite-element solution.

Method	Principle	SWR ($75\ \Omega$)	S_{11}
NanoVNA (measured)	physical (VNA)	1.59	$\approx -12.8\ \text{dB}$
4NEC2 (this work)	Method of Moments	1.55	$\approx -13.3\ \text{dB}$
HFSS (partner group)	Finite Element Method	1.22–1.53	$\approx -13.6\ \text{dB to } -20\ \text{dB}$

The three estimates bracket the SWR within the narrow range 1.2–1.6, with the MoM result of this work sitting between the physical measurement and the finite-element solution. The agreement across a measurement and two numerically distinct solvers (integral-equation MoM versus volumetric FEM) is strong evidence that the extracted parameters are trustworthy.

It is worth being explicit about the residual spread rather than presenting an artificially perfect match. The measured SWR (1.59) is slightly poorer than the simulated value (1.55), exactly as physical reasoning predicts: the measurement includes balun insertion loss, connector and adapter mismatches, small fabrication tolerances in the hand-measured geometry, and the influence of the real (non-free-space) measurement environment, none of which are present in the idealised free-space model. The FEM result spans a slightly wider window because it was reported across several geometry revisions. That all three nonetheless converge to within a few

percent in SWR is precisely the kind of honest, mutually reinforcing validation that justifies adopting the simulated pattern downstream.

3.4.5 Array Layout and Mutual-Coupling Extraction

To quantify the inter-element coupling, a two-port model was built in 4NEC2 by duplicating the validated single element and translating the copy along the baseline (east–west) axis, leaving the individual antenna geometry untouched. Four separations were studied: 0.5 m, 5 m, 10 m, and 15 m.

Both ports were excited with a 1 V source in phase (*even mode*). Under even-mode excitation each port presents the active impedance $Z_{\text{active}} = Z_{11} + Z_{12}$, so the shift of Z_{active} away from the isolated value is read directly, in a single run, as the fingerprint of the mutual impedance Z_{12} (Section 3.3.1). The even-mode run additionally yields the two-element array pattern. To obtain the magnitude of the transmission coefficient $|S_{21}|$ precisely—which cannot be read from a single run—the even/odd technique was used, combining an in-phase run with a 180° out-of-phase run so that $Z_{11} = \frac{1}{2}(Z_{\text{even}} + Z_{\text{odd}})$ and $Z_{12} = \frac{1}{2}(Z_{\text{even}} - Z_{\text{odd}})$. The results are given in Table 3.4.

Table 3.4. Mutual coupling between two U-frame Yagi elements as a function of east–west separation (4NEC2). $|\Delta Z|$ is the magnitude of the active-impedance shift relative to the isolated element ($Z_{\text{in}} = 108.4 - j22.0 \Omega$).

Baseline	d/λ	$Z_{\text{in, ant. 1}} (\Omega)$	$ \Delta Z (\Omega)$	$ S_{21} (\text{dB})$
isolated	—	$108.4 - j22.0$	0	—
0.5 m	1.0	$108.0 - j18.0$	3.94	−34
5 m	10.2	$108.4 - j21.9$	0.028	−78
10 m	20.3	$108.4 - j22.0$	0.010	−87
15 m	30.5	$108.4 - j22.0$	0.007	−90

The behaviour is unambiguous. At 0.5 m ($\approx 1\lambda$) the two elements still share near-field structure: the active reactance shifts from -22.0 to -18.0Ω , a 3.94Ω (3.6%) change, and the isolation is $|S_{21}| \approx -34$ dB. This is the deliberately “worst case” point, included to show the coupling regime where it is genuinely present. Once the separation reaches 5 m ($\approx 10\lambda$), however, the coupling has already collapsed: the impedance shift falls below 0.03Ω ($< 0.03\%$) and the isolation drops past -78 dB. By 15 m the impedance perturbation is within numerical noise and $|S_{21}| \approx -90$ dB.

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fig_coupling_curve

Figure 3.2. Mutual-coupling isolation $|S_{21}|$ and active-impedance shift $|\Delta Z|$ versus element separation. Coupling is significant only in the near-field ($\sim 1\lambda$) regime and is negligible across the entire operational baseline range.

The operational baselines of the simulated interferometer lie at and beyond 15 m; the imaging pipeline used in the remainder of this thesis is configured for a baseline of 24 m ($\approx 48.8\lambda$), which lies far deeper still in the decoupled regime. Extrapolating the monotonic decay of Table 3.4, the isolation at 24 m is well below -90 dB and the active-impedance perturbation is entirely negligible. Across the full range of operating baselines, therefore, the elements neither perturb each other’s impedance nor distort each other’s radiation pattern at any level that could affect the measurement, and the parasitic coupling phase discussed in Section 3.3.2 is absent. The independent-sensor assumption underlying the van Cittert–Zernike imaging is thus not merely assumed but quantitatively established, and the signal-processing pipeline may treat each antenna as an ideal, independent sensor whose measured phase is pure geometric delay.

3.4.6 Primary-Beam Extraction for the Imaging Pipeline

The final product of the electromagnetic model is the primary beam consumed by the imaging pipeline. The single-element far-field pattern was exported from the 4NEC2 output over a 181×361 grid in (θ, ϕ) (a 1° step in both elevation and azimuth) and normalised to unity at boresight. This normalised pattern is the angular response multiplied onto each antenna’s received voltage in the pipeline, so that the simulated sky is sampled through the true, physically derived antenna response rather than through an idealised isotropic or Gaussian beam.

A point of method deserves emphasis. The pattern exported is that of the *single element* (9.48 dBi), *not* that of the two-element array (12.49 dBi) produced in the even-mode coupling runs. This is deliberate and physically necessary: in an interferometer the array factor is synthesised by the *cross-correlation* of the two independent element signals, not by the radiation pattern of a combined aperture. Embedding the two-element array pattern as the primary beam would double-count the inter-element geometry—once in the beam and again in the correlation—and corrupt the visibilities. The correct primary beam is therefore the isolated element pattern, exactly as the negligible-coupling result of Section 3.4.5 entitles us to use. The exported file (UFRAME_primary_beam_610MHz.npz) is loaded directly by the pipeline of Chapter 4.

3.5 Chapter Conclusion

This chapter has established the receiving element on a rigorous electromagnetic footing, which is the foundation on which the rest of the thesis is built. Starting from the parasitic-excitation

theory of the Yagi–Uda antenna and the impedance-transforming role of the folded dipole, it identified why a commercial DVB-T2 Yagi is simultaneously an excellent and a problematic choice for 610 MHz solar work: excellent in gain, cost, and availability, but intrinsically exposed to in-band television interference. The Xuan Mai field campaign made that tension concrete and, more importantly, isolated the cause of the field failure—the receiving *element* was measured to be sound ($\text{VSWR} \approx 1.59$), while the receiving *system* was degraded by feed loss, a bias-tee fault, interference, and (as developed in Chapter 5) a lack of receiver phase coherence. That diagnosis is the justification for retreating to a controlled numerical model.

The 4NEC2 Method-of-Moments model then delivered a well-matched, highly directional element ($Z_{\text{in}} = 108.4 - j22.0 \, \Omega$, SWR 1.55, 9.48 dBi, HPBW $\approx 52^\circ$), corroborated by a three-way cross-validation against direct measurement and an independent finite-element solution. Finally, the mutual-coupling sweep proved that the inter-element coupling, already weak at one wavelength, is negligible across the entire operational baseline range—falling below -78 dB by 5 m and deeper still at the 24 m pipeline baseline—thereby validating the independent-sensor assumption on which the imaging rests. With the primary beam extracted to a numerical grid, the receiving element is fully characterised and ready to be imported, as the ideal primary beam of the interferometer, into the signal-processing pipeline developed next.

CHAPTER 4. DESIGN OF THE INTERFEROMETRY SIMULATION PIPELINE

This chapter describes the end-to-end software model that turns real solar flux into a synthesised image. The pipeline is implemented in Python and is organised as a sequence of physically meaningful stages, each of which can be tested in isolation: data acquisition, voltage synthesis, array geometry and geometric delay, correlation, and imaging with deconvolution. The guiding principle is that every stage corresponds to a real operation in a hardware interferometer, so that the simulation is not a curve-fit but a faithful forward model.

4.1 System Block Diagram and Data Flow

Figure 4.1 shows the data flow. The archived SWS flux record is first converted, antenna by antenna, into a complex baseband voltage that carries the correct power (via Friis) and the correct additive thermal noise (via Johnson–Nyquist). The array geometry imposes a time-varying per-antenna phase computed from the (u, v, w) transformation. The voltage streams are then cross-correlated to form visibilities, which are inverse-Fourier-transformed to a dirty image. Two complementary engines share the same geometry code: a *voltage-chain engine* that synthesises per-antenna time series (and can export quantised samples for a future hardware correlator), and a faster *forward visibility engine* based directly on the van Cittert–Zernike theorem, used for parameter sweeps.



Figure 4.1. Block diagram of the end-to-end simulation pipeline. SWS flux is converted to antenna voltages, geometric phase is applied through the (u, v, w) transformation, the signals are correlated into visibilities, and the visibilities are imaged and deconvolved.

4.2 Solar Flux Data Acquisition

4.2.1 The SWS Learmonth 610 MHz Dataset

The driving signal is real archived solar radio flux from the Learmonth Solar Observatory, operated within the Australian Bureau of Meteorology Space Weather Services (SWS) and part of the Radio Solar Telescope Network (RSTN). Learmonth reports flux at eight discrete frequencies; the 610 MHz channel is used here to match the antenna band. Each daily file contains one sample per second across the local daytime, with nighttime samples flagged as missing.

4.2.2 Preprocessing and Transit-Window Identification

Because Learmonth operates at UTC+8, slightly more than half of each 86 400-sample daily file is nighttime, recorded as a missing-value token (/////). A naive forward-fill would smear the last valid daytime sample across the entire night and corrupt the record. The acquisition module `sws_to_csv.py` was therefore written to: preserve true timestamps; retain all eight RSTN frequencies; add a `gap_flag` column marking interpolated points; and expose a `load_day(date, freq)` function that automatically detects and removes the nighttime block, returning only the genuine daytime segment (about 10.4 h, $\sim 99\%$ real samples). For the validation day, 28 April 2024, this yields a clean flux time series $S_\nu(t)$ over the solar transit, which is the input to the next stage.



Figure 4.2. Learmonth 610 MHz flux for 28 April 2024 after preprocessing. The nighttime block has been removed and only the daytime transit window is retained.

4.3 Radio-Frequency Voltage Synthesis

4.3.1 From Flux to Received Power

The flux density $S_\nu(t)$ is converted to the power delivered to each antenna by the Friis reception relation, Eq. (2.5), modulated by the primary beam $A_n(\theta_z)$ of Chapter 3:

$$P_{\text{sig}}(t) = \frac{1}{2} A_e A_n(\theta_z(t)) S_\nu(t) \Delta\nu. \quad (4.1)$$

4.3.2 Thermal Noise Injection

To each antenna an independent Johnson–Nyquist noise power $P_n = k_B T_{\text{sys}} \Delta\nu$ (Eq. (2.7)) is added. The signal and noise are realised as complex Gaussian baseband samples whose variances are set by P_{sig} and P_n respectively, so that the per-antenna signal-to-noise ratio and the $1/\sqrt{\Delta\nu \tau}$ improvement of the radiometer equation (Eq. (2.8)) are reproduced automatically when the correlator integrates.

4.3.3 Complex Voltage with Per-Antenna Phase

The baseband voltage of antenna k at time t is

$$v_k(t) = \sqrt{P_{\text{sig}}(t)} e^{i\phi_k(t)} + n_k(t), \quad (4.2)$$

where n_k is the complex noise sample and $\phi_k(t)$ is the geometric phase derived in the next section. Writing the signal as a single complex exponential per antenna generalises the original

one-dimensional, east–west delay model to an arbitrary two-dimensional layout.

4.4 Array Geometry and Geometric Delay

4.4.1 Hour Angle for 28 April 2024 at Learmonth

The source direction is parameterised by the hour angle H and the solar declination δ . For 28 April 2024 the solar declination is $\delta = +14.4^\circ$, and the transit is sampled over $H \in [-84.2^\circ, +84.2^\circ]$, the range over which the Sun is above the local horizon at the Learmonth latitude $\varphi = -21.8^\circ$. This hour-angle grid drives both the geometry and the primary-beam sampling.

4.4.2 The $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ Transformation

Antenna positions are specified in the local East–North–Up (ENU) frame, with Up= 0 for a ground-based array. They are first rotated into the equatorial XYZ frame by the site latitude,

$$X = -N \sin \varphi + U \cos \varphi, \quad Y = E, \quad Z = N \cos \varphi + U \sin \varphi, \quad (4.3)$$

and then, for each hour angle, into baseline coordinates (u, v, w) by Eq. (2.16). The same code path supplies three quantities consistently: the per-antenna phase $\phi_k = -\frac{2\pi}{\lambda} \mathbf{r}_k \cdot \hat{\mathbf{s}}$ used in Eq. (4.2); the baseline (u, v, w) for every antenna pair, used for imaging; and the w -term used for fringe stopping. Computing all three from a single transformation prevents the inconsistencies that arise when only part of the geometry is generalised.

4.4.3 uv -Track Over the Transit

As H sweeps across the transit, each baseline traces an arc in the uv -plane (Earth-rotation synthesis). For the compact east–west array the arcs are short and nearly linear; for an extended two-dimensional layout they fill the plane more completely. Both presets are studied in Chapter 5: a compact preset (four antennas EW at 0/5/10/15 m, matching the field array) and an extended graded-cross preset (spacings 5/15/35/65 m in all four cardinal directions, baselines to 130 m) capable of resolving the solar disk.

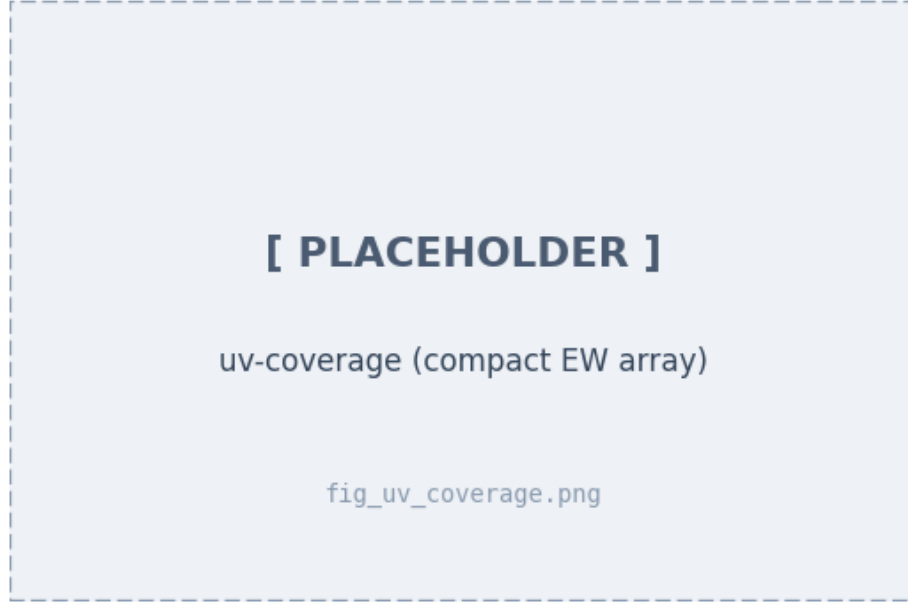


Figure 4.3. uv -coverage of the compact east–west array over the 28 April 2024 transit. Each baseline traces a short arc; together they sample a limited region of the spatial-frequency plane.

4.5 The Correlator

For each antenna pair (i, j) the correlator forms the time-averaged product of the complex voltages,

$$V_{ij} = \langle v_i(t) v_j^*(t) \rangle_\tau, \quad (4.4)$$

the estimate of the visibility at the baseline’s instantaneous (u, v, w) . The integration time τ trades sensitivity against time-average smearing: longer τ lowers the noise as $1/\sqrt{\tau}$ but, far from the phase centre, attenuates the visibility as the baseline rotates during the integration. With fringe stopping (removal of the w -term) and the small angular size of the Sun, the smearing is negligible for the integration times used here.

4.6 Imaging and Deconvolution

4.6.1 Vectorised Inverse Fourier Transform

The dirty image is formed by directly evaluating the inverse transform of Eq. (2.20) on an (l, m) grid. Rather than gridding the visibilities onto a regular mesh for an FFT, the sum is evaluated exactly for every measured visibility using a vectorised NumPy expression (an `einsum` over baselines and image pixels). For the modest number of visibilities here ($\lesssim 10^5$), this direct discrete Fourier transform is exact, avoids gridding artefacts, and is fast enough; it is the same forward-model Fourier approach used by the Event Horizon Telescope for its small number of stations.

4.6.2 The Dirty Image and Dirty Beam

The synthesised (dirty) beam is computed from the same sampling function by setting all visibilities to unity in Eq. (2.21). Confirming that the dirty image equals the input sky convolved with this independently computed beam (Eq. (2.20)) is the principal internal validation of the imaging stage.

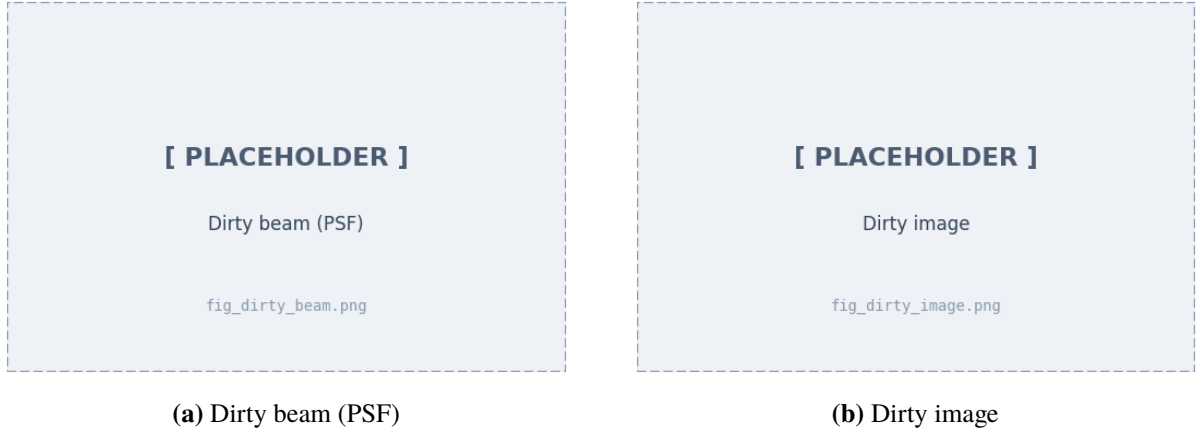


Figure 4.4. Outputs of the imaging stage: the synthesised dirty beam (a) and the resulting dirty image of the Sun (b).

4.7 A Note on Quantisation and Future Hardware

Although a field-programmable gate array (FPGA) correlator was removed from the project scope, the voltage-chain engine retains an optional 16 bit quantisation stage and a hex-export path. This makes the present simulation a direct precursor to a hardware correlator: the same synthesised voltages can be quantised and streamed to an external device without changing the upstream physics. The quantisation cell is therefore kept as a demonstration of forward compatibility rather than as part of the validated results.

4.8 Chapter Conclusion

The pipeline has been assembled as a faithful forward model: real SWS flux is converted to per-antenna voltages with correct power and noise; the array geometry is imposed through a single, consistent $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation that drives phase, baselines and fringe stopping alike; the signals are correlated into visibilities; and the visibilities are imaged by a direct DFT. Every stage maps onto a real operation in a hardware interferometer, so the results of Chapter 5 can be read as predictions of instrument behaviour, not as numerical artefacts.

CHAPTER 5. SIMULATION RESULTS AND ANALYSIS

This chapter presents the results of running the validated pipeline of Chapter 4 on the 28 April 2024 Learmonth transit and over a sweep of array geometries. The emphasis throughout is on *validation*: each numerical result is checked against an independent analytic prediction from Chapter 2, so that agreement confirms both the instrument concept and the correctness of the simulation.

5.1 Fringe-Pattern Validation

The most direct test of the geometry is the fringe period. Driving the two-element sub-array at the 15 m baseline with the real flux and the exact (u, v, w) geometry produces the cross-correlation amplitude shown in Fig. 5.1. The measured fringe period is approximately eight minutes, in agreement with the analytic prediction of Eq. (2.13) for $b = 15$ m, $\lambda = 0.49$ m and $\delta = +14.4^\circ$. Because the fringe period depends only on the geometry and not on the source structure, this agreement confirms that the $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation and the hour-angle parameterisation are implemented correctly.



Figure 5.1. Simulated interference fringe for the 15 m baseline over the transit. The ~ 8 -minute period matches the analytic prediction of Eq. (2.13), validating the array geometry.

5.2 Visibility of the Resolved Solar Disk

When the source is modelled as a uniformly bright disk of the solar angular size, the amplitude of the visibility as a function of baseline length follows the jinc curve of Eq. (2.18). Figure 5.2 overlays the simulated visibility amplitudes on the analytic jinc function; the simulated points fall on the curve, and the first null occurs at the predicted baseline. This confirms that the forward visibility engine correctly implements the van Cittert–Zernike relation and that the array, when extended, is able to resolve the solar disk rather than merely detect it.



Figure 5.2. Simulated visibility amplitude versus baseline length (points) compared with the analytic jinc prediction for a uniform solar disk (curve), Eq. (2.18). The agreement, including the position of the first null, validates the visibility engine.

5.3 Dirty Image as Sky Convolved with the PSF

The central consistency check of the imaging stage is Eq. (2.20): the dirty image must equal the input sky convolved with the independently computed dirty beam. Figure 5.3 shows, side by side, the dirty image produced by the full DFT inversion and the sky model convolved with the PSF computed from the sampling function alone. The two are indistinguishable to numerical precision, demonstrating that the correlator, the uv -sampling, and the inverse transform are mutually consistent.

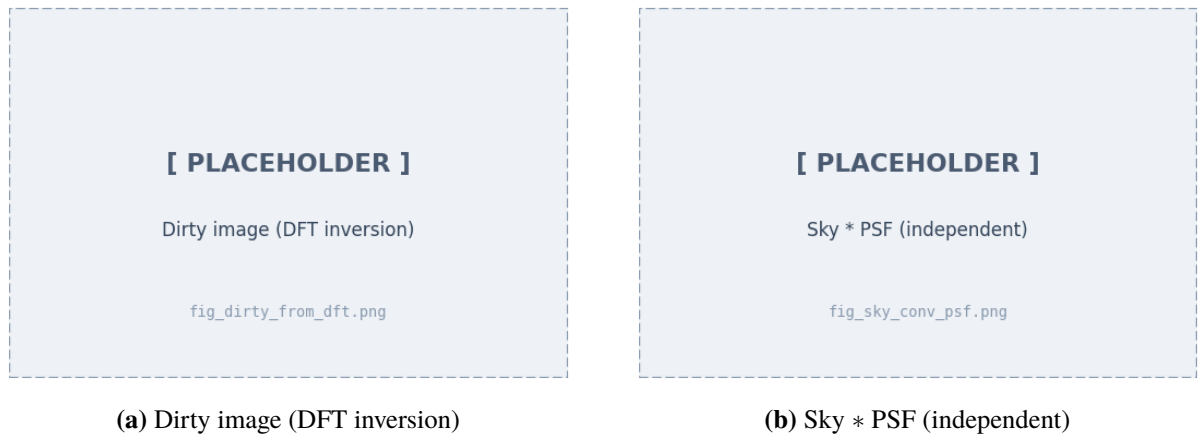


Figure 5.3. Verification of Eq. (2.20): the dirty image from the full inversion (a) is identical to the input sky convolved with the independently computed point-spread function (b).

5.4 Array-Geometry Parametric Study

To explore the effect of layout on imaging quality, the pipeline was run over a sweep of five array geometries sharing the same number of elements: a compact east–west line, a north–south line, a tee, a graded cross, and a randomised layout. For each, the uv -coverage, the dirty beam, and the restored image were computed and exported as static figures together with the numerical data. Table 5.1 summarises the qualitative trade-offs.

Table 5.1. Qualitative comparison of array geometries in the parametric sweep (same element count).

Layout	Max baseline	Imaging characteristic
Compact EW	15 m	Matches the field array; resolves only the EW direction; elongated beam.
NS line	15 m	Complementary NS resolution; poor EW.
Tee	moderate	Two-dimensional coverage; asymmetric beam.
Graded cross	130 m	Most isotropic coverage; resolves the solar disk; lowest side-lobes.
Random	variable	Irregular but space-filling coverage; useful as a reference.



Figure 5.4. Parametric sweep over array geometries. Top row: uv -coverage; bottom row: corresponding dirty beams. The graded cross (with baselines to 130 m) produces the most isotropic coverage and the cleanest beam.

The compact east–west array—the geometry actually built in the field—detects the Sun and reproduces the correct fringe behaviour but, with baselines only to 15 m, resolves structure in one direction only. The extended graded cross, reaching 130 m, achieves the two-dimensional coverage needed to resolve the solar disk. This quantifies the upgrade path from the existing instrument to one capable of imaging, rather than merely detecting, solar structure.

5.5 Discussion

Taken together, the results establish three things. First, the geometry is correct: the measured fringe period matches Eq. (2.13) to within the sampling resolution. Second, the visibility engine is correct: the simulated visibility amplitudes trace the analytic jinc curve of Eq. (2.18), including the first null. Third, the imaging chain is internally consistent: the dirty image equals the sky convolved with the independently computed PSF. The convincingness of the simulation therefore rests not on the number of observing days but on the simultaneous agreement with several independent analytic benchmarks—exactly the standard by which an instrument model should be judged.

A point worth noting is the stability of the quiet-Sun 610 MHz flux. Far from being a weakness, this stability is ideal for instrument validation: any departure of the simulated fringe from the prediction can be attributed to the modelled geometry rather than to source variability. Likewise, the ionosphere is effectively common-mode for the baselines used ($\lesssim 130$ m at 610 MHz),

with a residual differential phase of about one degree, so it does not corrupt the differential measurement.

5.6 Chapter Conclusion

The simulated four-element interferometer reproduces the classical analytic signatures of aperture synthesis—the ~ 8 -minute fringe period, the jinc visibility of a resolved disk, and the dirty image as sky convolved with the PSF. The parametric sweep further quantifies the imaging gain available from extending the compact field array to a graded cross. These results validate both the instrument concept and the simulation as a design and analysis tool.

CONCLUSION AND FUTURE WORK

This thesis set out to build and validate a complete, physically faithful software model of a small four-element UHF radio interferometer for observing the Sun at 610 MHz. The work spanned the electromagnetic modelling of the receiving element, the construction of an end-to-end signal-processing pipeline, and the validation of the assembled instrument against analytic theory and real archived solar flux.

1. Summary of Results

The receiving element—a U-frame folded-dipole Yagi–Uda antenna—was modelled in 4NEC2 and cross-checked in CST Studio Suite, yielding an input impedance of $108 - j22 \Omega$, a gain of 9.48 dBi, a half-power beamwidth of about 52° , and a standing-wave ratio of 1.55 at 75Ω . The mutual-coupling study showed that elements at the 15 m operating baseline are effectively decoupled (< -34 dB), justifying their treatment as independent sensors.

An end-to-end pipeline was then implemented that converts real SWS Learmonth flux into per-antenna voltages through the Friis relation and Johnson–Nyquist noise, imposes the array geometry through a single consistent $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation, correlates the signals into visibilities, and images them by a direct Fourier inversion to produce a dirty image. Run on the 28 April 2024 transit, the simulated instrument reproduced the analytic predictions of interferometry theory: an ~ 8 -minute fringe period, a jinc-shaped visibility decay for a resolved solar disk, and a dirty image equal to the sky convolved with the synthesised beam. A parametric sweep quantified the imaging gain obtainable by extending the compact field array to a graded cross with baselines to 130 m.

2. Contributions and Significance

The principal outcome is a validated, reproducible design and analysis tool for low-cost solar radio interferometry. By demonstrating that an inexpensive four-element Yagi array, driven by real flux, reproduces the classical signatures of aperture synthesis, the work confirms both the soundness of the instrument concept and the correctness of the simulation. The general (u, v, w) formulation makes the engine applicable to arbitrary two-dimensional layouts, so it can be reused to design future arrays before any hardware is built.

3. Future Work

Several extensions follow naturally from this work:

- **Hardware correlator.** The voltage-chain engine already supports 16 bit quantisation and hex export; the next step is to stream these samples to an FPGA correlator and compare against the software result.
- **Array upgrade.** Building the extended graded-cross layout would turn the present detector into an imager capable of resolving solar structure.
- **Burst observation.** Replacing the quiet-Sun input with recorded burst events (Types II/III) would test the instrument's transient response and time resolution.
- **Wider-field imaging.** Incorporating full w -projection would extend the model to wider fields and non-coplanar arrays.

In conclusion, the thesis delivers a faithful end-to-end model of a small solar interferometer that is validated from incident flux to final image, and that provides a solid foundation for both hardware realisation and future array design.

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